SCHOOL OF COMPUTER ENGINEERING

III SEMESTER B.TECH.

Midterm Exam

SUBJECT: DIGITAL SYSTEMS & COMPUTER ORGANIZATION [CSE 2104]

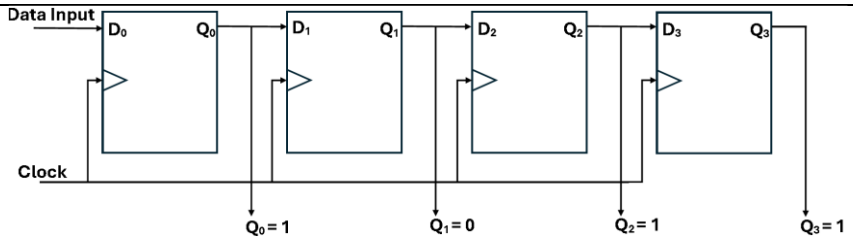
Time: 10.45-12.15 Date:16/9/25 MAX.MARKS: 30

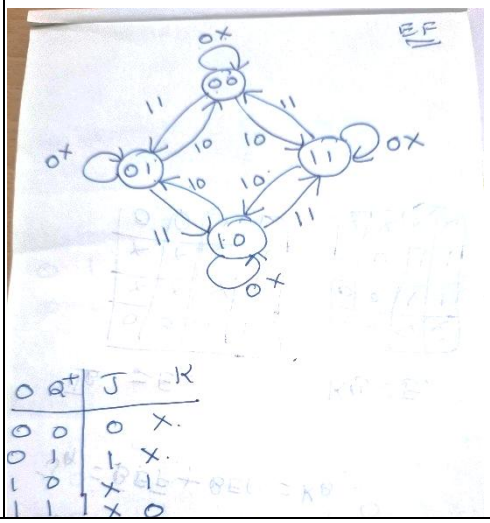
Note: Missing data may be assumed suitably.

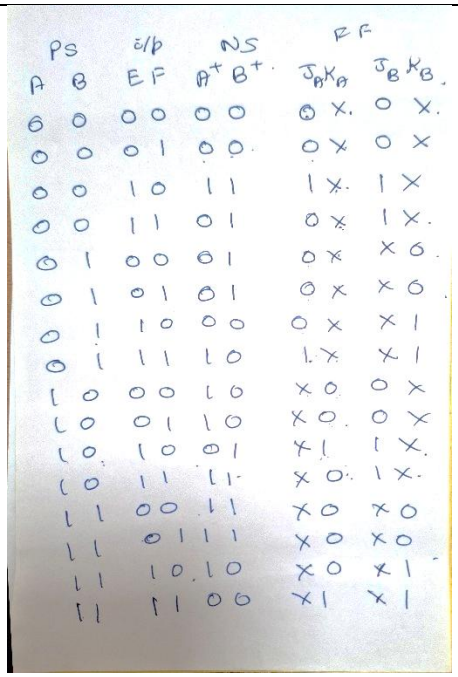
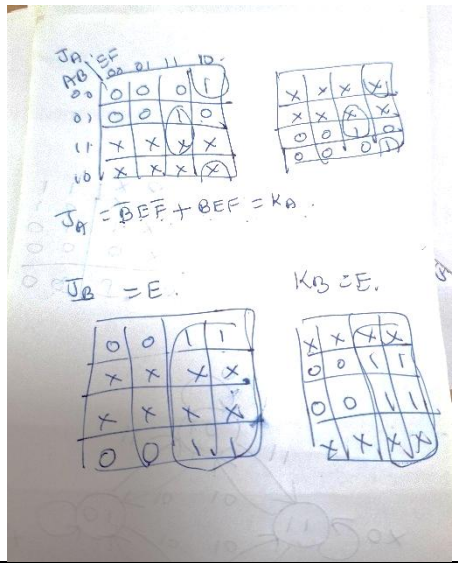
<table>
<tr><th>Q. No</th><th>Questions</th><th>Marks</th></tr>
<tr><td>1.</td><td>What is the minimum number of 8-to-1, 4-to-1, and 2-to-1 multiplexers required to implement a 32-to-1 multiplexer?
A) 4 of 8-to-1, 1 of 4-to-1, 0 of 2-to-1
B) 2 of 8-to-1, 2 of 4-to-1, 1 of 2-to-1
C) 8 of 4-to-1, 2 of 2-to-1
D) 1 of 8-to-1, 4 of 4-to-1, 1 of 2-to-1</td><td>0.5</td></tr>
<tr><td>2.</td><td>What is the output of a 3-to-8 decoder when the input is 101?
A) Output line 5 is high
B) Output line 2 is high
C) Output line 6 is high
D) Output line 7 is high</td><td>0.5</td></tr>
<tr><td>3.</td><td>In a priority encoder, if inputs D3 = 1, D2 = 1, D1 = 0, D0 = 1, what will be the output?
A) 00
B) 01
C) 10
D) 11</td><td>0.5</td></tr>
<tr><td>4.</td><td>Which of the following is used to detect overflow in binary addition?
a) Carry into MSB
b) Carry out of MSB
c) XOR of carry into and carry out of MSB
d) AND of MSB bits</td><td>0.5</td></tr>
<tr><td>5.</td><td>Which of the following is a valid way to declare a 4-bit register in Verilog?
a) reg[4] a;
b) reg[3:0] a;
c) register[3:0] a;
d) wire[4:1] a;</td><td>0.5</td></tr>
<tr><td>6.</td><td>A SR latch is

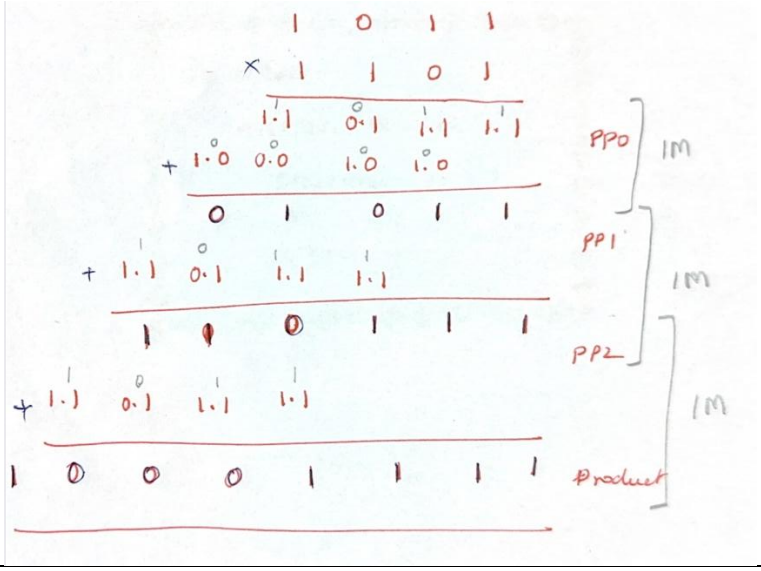
a) Combinational circuit
b) Synchronous sequential circuit
c) 1-bit memory element
d) 1-clock delay element</td><td>0.5</td></tr>
<tr><td>7.</td><td>The truth table of a MN – flipflop is given below:

<table><tr><td>M</td><td>N</td><td>Output</td></tr><tr><td>0</td><td>0</td><td>Toggle</td></tr></table></td><td>0.5</td></tr>
</table>

		0	1	1
		1	0	0
		1	1	No change
	The characteristic equation of MN- flipflop is: a) $Q(t+1) = MQ^I + N^IQ$ **b) $Q(t+1) = M^IQ^I + NQ$** c) $Q(t+1) = MQ + N^IQ^I$ d) $Q(t+1) = M^IQ+NQ^I$			
8.	Identify the TRUE statement/s with respect to Verilog code from the following: i) The procedural statements should appear inside the always block ii) A sub-circuit can be instantiated inside the always block iii) A subcircuit cannot be instantiated if the order of signals listed in the instantiation statement is not same as the order of the ports given in the module statement of the subcircuit 1. i) and ii) only 2. iii) only **3. i) only** 4. ii) and iii) only			**0.5**
9.	 The above figure depicts a 4-bit Serial-in-Parallel-out (SIPO) shift register that currently holds the value 1101. Q_3 is the Most Significant Bit (MSB) and Q_0 is the Least significant Bit (LSB). If a new value 0110 is supplied serially through the *Data Input*, what will be the content (as Q_3 Q_2 Q_1 Q_0) of the above register after three consecutive clock cycles? (Note: the serial input starts with the MSB and ends with the LSB of 0110) **1. 1011** 2. 1101 3. 0111 4. 0110			**0.5**
10.	Assume a 4-bit Johnson counter holds an initial value of 1000. Which of the following depicts the behaviour of a Johnson counter after applying three consecutive clock signals (Clk)? 1. Initial 1 0 0 0 Clk1 0 0 0 1 Clk2 1 0 0 0 Clk3 0 0 0 1			**0.5**

	2. Initial 1 0 0 0 Clk1 0 1 0 0 Clk2 0 0 1 0 Clk3 0 0 0 1 **3. Initial 1 0 0 0** **Clk1 1 1 0 0** **Clk2 1 1 1 0** **Clk3 1 1 1 1** 4. Initial 1 0 0 0 Clk1 0 1 0 0 Clk2 1 0 0 0 Clk3 0 1 0 0	
11.	Derive the expressions for the flip flop inputs of a sequential circuit with two *JK* flip-flops *A* and *B* and two inputs *E* and *F* . If *E* = 0, the circuit remains in the same state regardless of the value of *F* . When *E* = 1 and *F* = 1, the circuit goes through the state transitions from 00 to 01, to 10, to 11, back to 00, and repeats. When *E* = 1 and *F* = 0, the circuit goes through the state transitions from 00 to 11, to 10, to 01, back to 00, and repeats.  -1-	**4**

	 -1-  -2-	
12.	Multiply $(1011)_2$ and $(1101)_2$ using an array multiplier approach. Show the step by step operation to generate the final product.	**3**



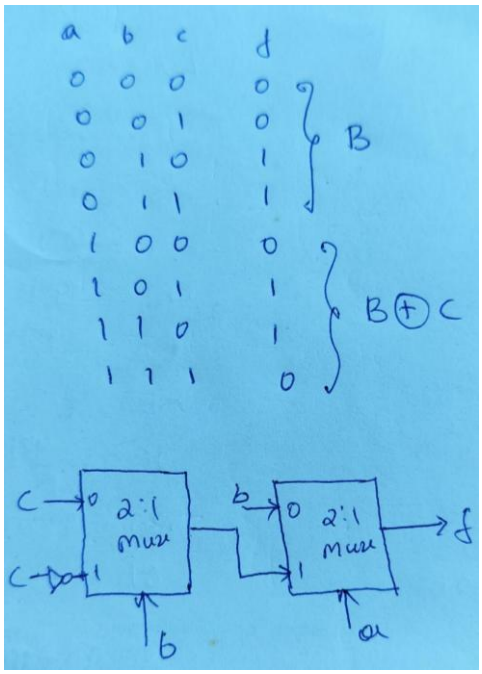
13. Write Verilog code for 2 to 1 multiplexer using conditional operator. Design f(a, b, c)=∑(2,3,5,6) using two 2 to 1 multiplexers and one inverter.

```
module mux2to1 (w0, w1, s, f);
input w0, w1, s;
output f;
assign f = s ? w1 : w0;
endmodule
```

OR

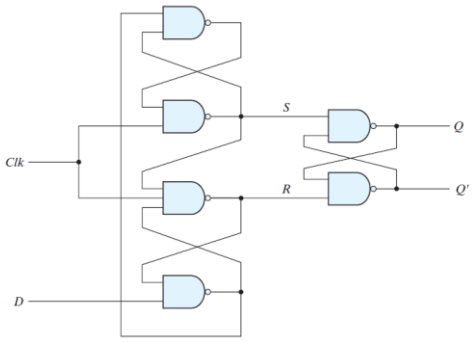
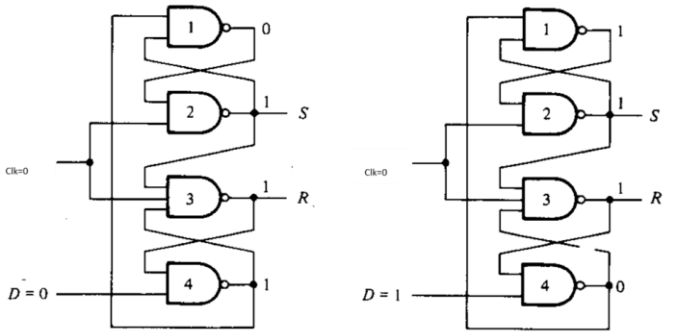
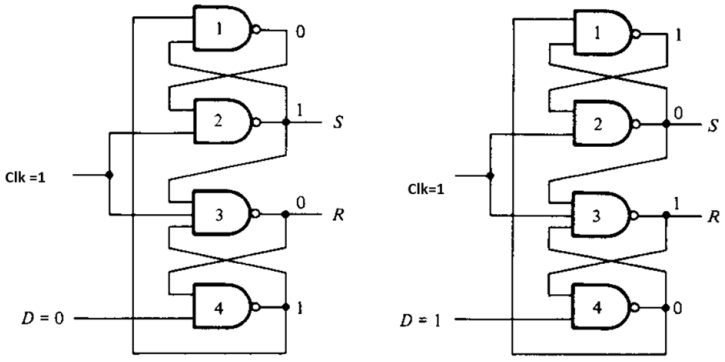
```
module mux2to1 (w0, w1, s, f);
input w0, w1, s;
output f;
reg f;
always @(w0 or w1 or s)
f = s ? w1 : w0;
endmodule
```

1M



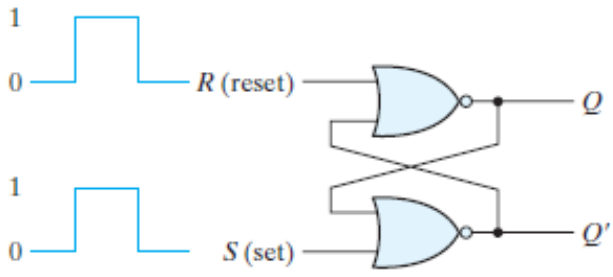
2M

3

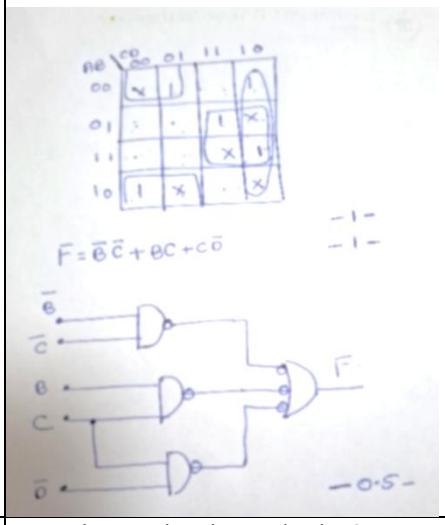
14.	Explain the working of edge triggered D Flip Flop with three crosscoupled NAND latches with a neat diagram and all the four cases of the inputs D and Clock  **1M**  (a) With Clk=0  (b) With Clk=1 **1.5M**	**3**

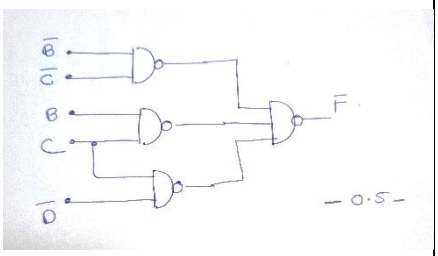
S	R	Q	Q′	
1	0	0	1	
1	1	0	1	(after $S = 1, R = 0$)
0	1	1	0	
1	1	1	0	(after $S = 0, R = 1$)
0	0	1	1	(forbidden)

(b) Function table

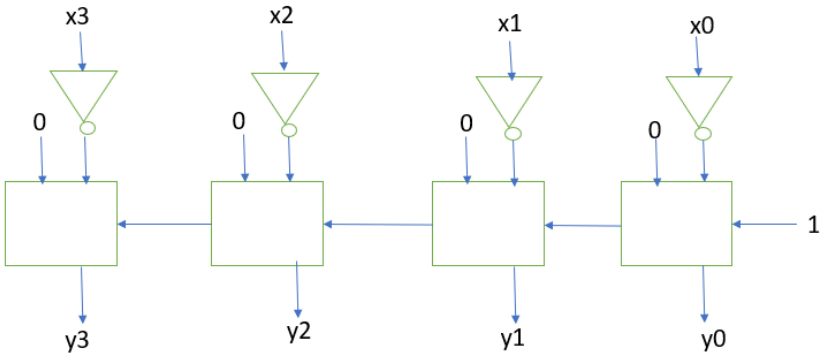
	• When **Clk = 0**, **S and R = 1**, so output holds its state (D may be 0 or 1). • When **Clk = 1** : • If **D = 0**, R → 0 → flip-flop resets (Q = 0). Further changes in D while Clk = 1 have no effect. • If **D = 1**, S → 0 → flip-flop sets (Q = 1). Again, changes in D while Clk = 1 do not affect output. • When **Clk returns to 0**, S and R go back to 1, latch is quiescent, output unchanged. • Flip-flop responds **only to 0 → 1 transition of Clk**; not to falling edges or steady levels. • **Setup time**: D must remain constant before Clk ↑ (equals delay of gates 4 and 1). • **Hold time**: D must remain stable after Clk ↑ (equals delay of gate 3 if D=0, or gate 2 if D=1). **Propagation delay**: time from trigger edge to stable output. **0.5M**	
15.	Explain why the SR NAND latch is called an active-low latch, and also draw the circuit diagram and truth table of an SR latch using NOR gates. In the NAND-based SR latch, the inputs are usually labeled S (Set) and R (Reset). When S (active-low) = 0 and R = 1, the output becomes high (Q = 1). Similarly, when S = 1 and R = 0, the output becomes low (Q = 0). Since the latch responds to a low input to activate the set or reset operation, it is called an active-low latch. 1M  (a) Logic diagram 1M S R | Q Q' 1 0 | 1 0 0 0 | 1 0 (after $S = 1, R = 0$) 0 1 | 0 1 0 0 | 0 1 (after $S = 0, R = 1$) 1 1 | 0 0 (forbidden) (b) Function table 1M	**3**
16.	Design a minimal NAND–NAND circuit realization for the expression, assuming that the variables are available in both complemented and uncomplimented form.	**3**

$$F(A,B,C,D) = \sum m(1,2,7,8,14) + d(0,6,9,10,15)$$





17. Design a circuit to obtain 2's complement of a 4 bit number using full adders and inverters. **2**

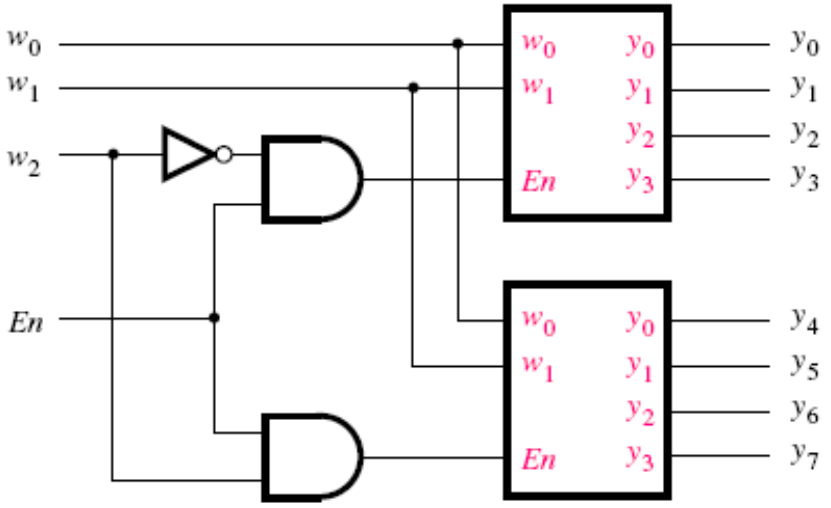


y3y2y1y0 is the 2's complement of X

18 Wrtite the truth table for 8 -to-3 priority encoder. Assume that the input is W[7:0] with W[0] having highest priority **2**

W 7	W 6	W 5	W 4	W 3	W 2	W 1	W 0	Y2	Y1	Y0	Z
X	X	X	X	X	X	X	1	0	0	0	1
X	X	X	X	X	X	1	0	0	0	1	1
X	X	X	X	X	1	0	0	0	1	0	1
X	X	X	X	1	0	0	0	0	1	1	1
X	X	X	1	0	0	0	0	1	0	0	1
X	X	1	0	0	0	0	0	1	0	1	1
X	1	0	0	0	0	0	0	1	1	0	1
1	0	0	0	0	0	0	0	1	1	1	1
0	0	0	0	0	0	0	0	X	X	X	0

19 Illustrate the method of combining two 2-to-4 decoders to design a 3-to-8 decoder. **2**



1.5M for Diagram

A 3-to-8 decoder is built using two 2-to-4 decoders with enable. Inputs w_0, w_1 go to both decoders, while w_2 selects which decoder is active through the enable logic (inverter + AND).

If $w_2 = 0$, upper decoder active → outputs y_0–y_3.

If $w_2 = 1$, lower decoder active → outputs y_4–y_7.

Thus, exactly one output (y_0–y_7) goes high for each input combination.

Explanation: 0.5M